Exploring digital competence's impact on faculty well-being through SEM-SVM analysis

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Abstract: *This research aims to explore the impact of digital teaching competence on faculty career well-being, teacher self-efficacy, and the moderating role of pedagogical content knowledge within the context of colleges in Pakistan. A quantitative approach was adopted, involving 266 faculty members from various academic programs in colleges across Pakistan. Data were collected through an online survey using validated scales adapted from prior research. The study used partial least squares structural equation modeling (PLS-SEM) to analyze the relationships between constructs. Support vector machine (SVM) analysis was conducted using JASP software to enhance predictive accuracy and identify the importance of key features in the dataset. The results confirm that digital teaching competence significantly impacts faculty career well-being and teacher self-efficacy. Teacher self-efficacy mediated the relationship between digital teaching competence and faculty career well-being, providing deeper insights into this linkage. Furthermore, pedagogical content knowledge moderated the effect of digital teaching competence on faculty career well-being, emphasizing its critical role. The inclusion of SVM analysis revealed the significance of variables such as psychological factors and AI exposure in influencing predictive performance, further validating the robustness of the findings. This study contributes to the existing literature by demonstrating the critical role of digital teaching competencies in enhancing faculty members' well-being and effectiveness. It underscores the need for targeted professional development programs that integrate technology and pedagogy, ultimately fostering a more supportive educational environment for faculty and improving overall teaching quality.*

Keywords: *Digital teaching competence, teacher self-efficacy, pedagogical content knowledge, faculty career well-being, support vector machine.*

JEL Classification: *I23, J24, O33.*

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Introduction

The integration of digital technologies in education has transformed faculty roles in institutions of higher learning, and new teaching and learning practices become more prominent. Effective pedagogy requires a faculty member who is competent in digital teaching, which enables him to use technology to engage and facilitate student learning outcomes (Yang & Du, 2024). The shifting educational landscape has made it essential to respond to the plethora of needs that have emerged in response to digital education; faculty members' digital competencies are among them (Nousheen et al., 2024). The transformation thus calls for an inclusive comprehension of how digital teaching competency affects the various aspects of educators' professional lives (Kruskopf et al., 2024), such as well-being and self-efficacy (Martinsone & Vanaga, 2024). Therefore, studying the relationship between digital teaching competence, teacher self-efficacy, and faculty career well-being is essential in addressing the challenge of modern education.

Research has described the interaction between digital teaching competence, teacher self-efficacy, and faculty well-being (Khan et al., 2022; Valle et al., 2024). For instance, it has been found that educators who possess digital literacy and competency in digital tools have a higher sense of self-efficacy, hence enhancing their instructional delivery (Valle et al., 2024). In addition, research suggests that higher self-efficacy is positively associated with faculty career well-being because the faculty's confidence in their employability skills will increase their affection for their job (Orakova et al., 2024). In addition, different findings have been made that digital teaching competency may lead to better pedagogical strategies and, therefore, contribute to an effective supportive learning climate that increases the professional satisfaction of educators (Lee et al., 2024). In light of this, the studies outline how digital competency impacts educators' career experiences and outcomes (Leino et al., 2024). Moreover, research studies indicate that faculty members who are highly digitally literate have better command over their workplace and are more adaptable to the nature of changing educational settings, which may lead to better career well-being (Gordon et al., 2024). This literature is of the same sort where the authors highlight the importance of digital skill development to safeguard professional identity and mitigate burnout among the staff (Nissim & Simon, 2024).

The complete understanding of the effects of digital teaching competence on faculty careers, however, is limited by the gap in the literature (Boeve-De Pauw et al., 2024). There needs to be more research investigating potential interactions between teacher self-efficacy or pedagogical content knowledge and digital teaching competence about faculty outcomes (Ghazali et al., 2024; Khan et al., 2022). Although the benefits of digital competence per se are as evident as the daylight, digital competence's role in moderating or mediating other variables in influencing faculty well-being remains unexplored (Täschner et al., 2024). Also, most publications focus on elementary or secondary school teachers, while college and university teachers hardly receive attention (Liu et al., 2024). Such a vast gap persists about how college faculty members manage challenges from digital teaching and how this would affect their job satisfaction (Šimunović et al., 2024). Additionally, few studies have used longitudinal research to analyze the long-term effects of digital competencies on faculty's well-being, thus limiting the extent to which time-evolving changes in the impact of digital literacy can be captured (Ninković et al., 2024). These gaps would be filled so faculty career well-being and digital teaching competency could better understand the subtle interrelationships between these factors (Ali et al., 2024).

To establish a credible motivation towards this research, there is a need to discover the growing importance of digital competence in higher education, particularly regarding the driving forces of accelerated technological change and pedagogical innovation. This drift towards digital learning environments has raised unprecedented challenges for academics to develop digital teaching competence to maximize their performance and overall career well-being. Despite increasing interest in the digitalization of education, a critical knowledge gap exists regarding the impact of digital teaching competence on faculty career wellness and self-efficacy. Theoretical frameworks provide a foundation for this research. Bandura's social cognitive theory focuses on the role of self-efficacy in triggering behaviors, suggesting that staff with greater digital capabilities will be more likely to possess higher self-efficacy

and, consequently, improved career success (Kholifah et al., 2023). The technology acceptance model also focuses on perceived ease of use and usefulness as the primary drivers of technology adoption. Faculty members who view digital teaching tools as helpful are more likely to use them successfully in their pedagogy, which contributes to greater job satisfaction and career well-being (Wang & Chu, 2023).

Based on these theoretical foundations, this research aims to address significant research gaps by investigating the impact of digital teaching competency on faculty career happiness and self-efficacy. Moreover, it also explores teacher self-efficacy as a mediating variable and pedagogical content knowledge as a moderator of this relationship. As opposed to past studies isolating these variables, this paper is an interdisciplinary study that helps in understanding how these relationships take place in a university setting. Through this process, it provides new insights about how faculty members can leverage digital competencies so as to grow more professionally engaged and productive. Secondarily, the paper contributes to general technology-enhanced learning literature by grounding faculty development strategies by institutions with empirical evidence. To ensure fundamental clarity and coherence, the paper is organized as follows: after the introduction is a review of theory and methodology germane to the research, followed by methodology outlining data collection and analysis processes. It is followed by a section of findings, which are then interpreted within the framework of theory and practice implications. Lastly, the research ends with future research directions and limitations, presenting potential avenues for further research in this critical field.

1 Theoretical background

Education for well-being emphasizes the holistic development of students, focusing not only on academic achievements but also on emotional, social, and mental health aspects (Wong et al., 2023). Research highlights faculty members' pivotal role in shaping this comprehensive educational experience (Lekhu, 2023). Faculty are increasingly recognized as key facilitators in promoting well-being through direct instruction and creating supportive learning environments (Uçar & Zarfsaz, 2023). Studies have shown that when educators incorporate well-being frameworks into their teaching, students exhibit higher engagement, resilience, and self-awareness levels. These frameworks often promote positive relationships, mindfulness, emotional intelligence, and problem-solving skills (Zheng et al., 2023). Faculty members can help cultivate a culture of care and empathy through their interactions and instructional strategies, which is crucial for students' development. This enhances student outcomes and fosters a positive educational climate that benefits learners and educators (Göbel et al., 2023).

Additionally, research has highlighted the importance of faculty well-being in achieving these educational goals (Su, 2023). Educators who experience high levels of job satisfaction, emotional support, and professional development opportunities are more likely to engage in practices that promote student well-being (Chen, 2023). Faculty well-being directly influences their teaching effectiveness, ability to connect with students, and resilience in managing classroom challenges. Educational institutions prioritizing their faculty's well-being often see a trickle-down effect, where empowered and supported teachers create more inclusive and supportive classrooms (Sehar & Alwi, 2023). Literature in this domain underscores the need for systemic changes, advocating for policies integrating well-being into educational curricula and providing continuous support for educators (Rastegar & Rahimi, 2023). This dual focus on faculty well-being and the well-being of students forms the cornerstone of a sustainable and nurturing educational ecosystem.

Therefore, studies have found that digital teaching capabilities are an essential predictor of faculty career well-being (Hershkovitz et al., 2023). The research has established that such faculties would enjoy greater job satisfaction, less stress, and higher overall well-being with greater digital teaching capabilities (Guoyan et al., 2023). For instance, research has found that digitally literate teachers often have a positive attitude toward their teaching context and are much more actively engaged in their practice (Masoumi & Noroozi, 2023). Faculty members who use technology effectively in their teaching practices are most frequently characterized by much higher job satisfaction and improved well-being (Shu, 2022). In addition, studies have shown a relationship between digital competence and professional development opportunities, so it is more likely

that these faculties who invest time in developing their digital competencies will have their career paths well-supported (Liang et al., 2022). Based on these findings, digital teaching competence is a crucial factor influencing faculty career well-being (Shonfeld et al., 2022). The improving digital competencies for educators help them to develop more security and self-confidence in their pedagogy, and changes in the educational landscape become more amendable for them (Liu et al., 2022). Indeed, some of these changes can alleviate inadequacy and frustration among educators needing help using digital materials (Zhang & Fang, 2022). Empirical evidence also suggests that developing digital competencies enhances teaching quality and a collaborative, innovative educational climate, which maximizes welfare among faculty staff (Yang & Du, 2024). It is a reasonable suggestion to advance that higher digital teaching competence positively correlates with the better career well-being of faculty members.

H1: Digital teaching competence significantly influences the faculty's career well-being.

However, many scholars have researched the effects of digital teaching competency on a teacher's self-efficacy, focusing only on how technology shapes an educator's confidence in performing his or her job (Kruskopf et al., 2024). According to studies, teachers with more experience working with digital tools display relatively higher self-efficacy beliefs in the probability of effective instruction and interaction with students (Valle et al., 2024). For example, research findings showed that training and resource aid for instructors to develop their digital skills enhance the self-efficacy level of instructors in instructional effective practices and subsequent learning results (Lee et al., 2024). Empirical evidence also proves that digitally more competent teachers are typically intrinsically innovative in educational change, which drives and verifies their self-efficacy belief in being a good teacher (Gordon et al., 2024). Having established the nexus between digital teaching competency and self-efficacy, it becomes feasible to hypothesize that there is a significant relationship between the two constructs (Boeve-De Pauw et al., 2024). The more the teacher develops his digital skills, the more adept he becomes with different teaching situations and thus would be imbued with mastery over his teaching practice (Täschner et al., 2024). This mastery will likely be translated into improved self-efficacy, allowing the teacher to exercise higher confidence when discharging his or her role (Šimunović et al., 2024). In addition, empirical research points out that instructor self-efficacy goes hand in hand with instructional quality and, therefore, in any case, attests to the fact that developing digital competence would be beneficial not only to individual instructors but also to enhance student learning experiences (Ali et al., 2024). At this point, it can be feasible to point out that instructional self-efficacy relies significantly on digital teaching competence.

H2: Digital teaching competence significantly influences teacher self-efficacy.

Some research has been directed to the mediating effect of instructor self-efficacy on learning outcomes (Wang & Chu, 2023). Research studies have shown that self-efficacy facilitates turning skills and competencies into positive career experiences (Lekhu, 2023). Research findings on digital teaching competencies indicate that teachers with high self-efficacy will facilitate better and more effective implementation of technology in class, significantly improving teaching practices and satisfaction with their careers (Lekhu, 2023). Empirical results indicate that self-efficacy influences educators' resilience and adaptive strategies to overcome digital instruction challenges (Zheng et al., 2023). The interplay points out that self-efficacy is not a byproduct of digital competence but rather a mediating variable of crucial influence on the relationship between digital teaching skills and overall career well-being (Su, 2023). This relationship, found in established studies between digital teaching competence and teacher or faculty career well-being self-efficacy, makes it reasonable to propose that self-efficacy significantly mediates the relationship between digital teaching competence and faculty career well-being (Hershkovitz et al., 2023). As educators develop digital skills, their self-efficacy will probably increase, positively affecting their career well-being (Masoumi & Noroozi, 2023). Empirical evidence indicates that teachers with high self-efficacy are more challenging to face, look for professional development opportunities, and have a positive future vision for their careers (Liang et al., 2022). Therefore, it can be an agency connecting the development

of digital teaching competency to faculty career well-being, underlining why it is very relevant in this relation.

H3: Teacher self-efficacy significantly mediates the relationship between digital teaching competence and the faculty's career well-being.

It has been widely realized through research that pedagogical content knowledge PCK is a significant part of effective pedagogical practices (Liu et al., 2022). Research reveals that teachers who possess a proper idea about pedagogy and content are much more capable of embedding digital learning tools in the learning process (Nousheen et al., 2024). This integration will facilitate more profound learning experiences to enhance faculty career well-being (Martinsone & Vanaga, 2024). For example, empirical evidence indicates that such teachers, being more PCK-oriented, are likely to use technology in proper educational ways, yielding healthier results for students and improved job satisfaction for teachers (Orakova et al., 2024). Moreover, research evidence has revealed that PCK is an important factor in determining teachers' resistance, readiness, and adaptation to new technologies, and these can, in turn, further enhance teachers' confidence and well-being within jobs (Leino et al., 2024). Based on such findings, one can hypothesize that pedagogical content knowledge significantly moderates the relationship between the level of digital teaching competence and the extent of faculty career well-being (Nissim & Simon, 2024). Faculties with high levels of PCK are likely to use digital competencies much more effectively, improving their practice and leading to better career experiences (Ghazali et al., 2024). This would imply that, in theory, the better the educators' digital skills and more profound pedagogical knowledge, the better the environments they would create, which favor well-being (Liu et al., 2024). Moreover, PCK may act as a buffer against all the digital instruction stressors, allowing faculty members to tackle the challenges more resiliently (Ninković et al., 2024). This leads to the plausibility of stating that PCK plays a critical moderating role in the relationship between digital teaching competence and faculty career well-being, modifying how educators experience their professional roles in this technology-rich landscape.

H4: Pedagogical content knowledge significantly moderates the relationship between digital teaching competence and teacher self-efficacy.

The proposed model of the study was derived from the previous literature, and shown in Fig. 1.

2 Research methodology

2.1 Research design

The study used a quantitative research design to examine the interrelationship between

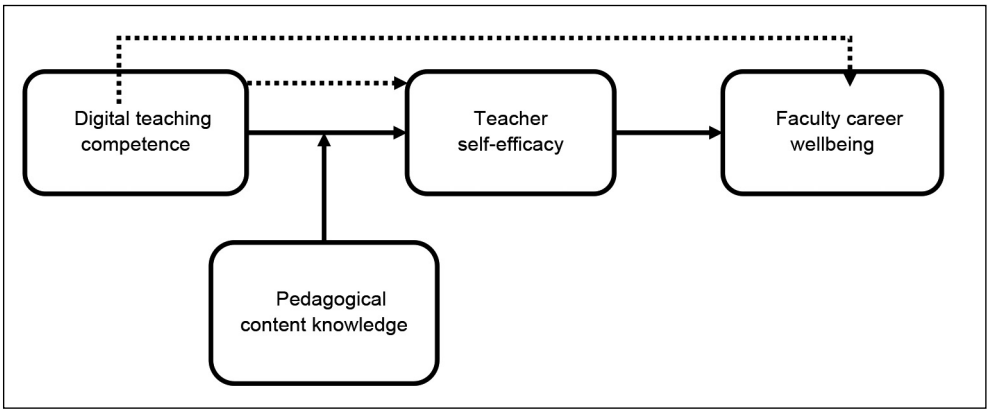


Fig. 1: Proposed model of the study

Source: own

digital teaching competence, lecturers' career well-being, teachers' self-efficacy, and pedagogical content knowledge about colleges within Pakistan. By applying established scales for measuring such constructs, this study provided empirical insight into how digital competencies impact lecturers' perceptions of their overall well-being and self-efficacy. Participants: this paper targeted lecturers who teach at various academic programs in colleges in various parts of Pakistan. A total of 266 respondents participated in this research. Stratified random sampling, whereby fair representations from various disciplines and different programs within the colleges were ensured, was adopted to select participants. An online questionnaire was sent through email and academic networks to involve everyone. The survey tool aimed to gather information regarding digital teaching competence, career well-being, self-efficacy, and pedagogical content knowledge regarding the faculty. The questionnaire included demographic questions to gather fundamental participant characteristics, such as age, gender, academic rank, and the years experienced in teaching.

2.2 Scales of measurement

The measurement scales used for this study were adopted from previous empirical research work for reliability and validity. Items that were used to measure established constructs were of proven psychometric properties. Digital teaching competence was measured through items that would evaluate the proficiency of faculty members with digital tools and technologies used in their teaching work. Faculty career well-being would be measured through items meant to capture job satisfaction, work-life balance, and overall professional fulfillment. Teacher self-efficacy was measured by faculty responses regarding the level of confidence attached to teaching skills and the ability to improve engagement. Lastly, pedagogical content knowledge was assessed using items mainly focused on integrating subject matter knowledge with practical strategies for teaching. Each item received ratings from a Likert scale that ranged from 1, where the respondents strongly disagreed with the given statement, to 5, which reflected an intensely agreeable response. Tab. 1 shows the items of questionnaire and measurements.

Tab. 1: Questionnaire and measurements

Construct	Items	References
Digital teaching competence	Five	Chu et al. (2023)
Teacher self-efficacy	Thirteen	Liu et al. (2023)
Pedagogical content knowledge	Six	Mensah (2023)
Faculty career well-being	Eight	Liu et al. (2023)

Source: own

2.3 Estimation techniques

Partial least squares structural equation modeling (PLS-SEM) is a robust statistical technique for examining complex relationships between variables. PLS-SEM was chosen because it can also use relatively small samples and provide insights into the measurement and structural models (Ahmed et al., 2024). The procedure adopted in this study consisted of two primary stages: the initial assessment of the measurement model and the structural model's evaluation. Measurement model – the reliability and validity of the scales were considered. Reliability assessment was taken from Cronbach's alpha and composite reliability coefficients, while the measures that follow established validity criteria were presented with average variance extracted (AVE). Discriminant validity was measured using the Fornell-Larcker criterion and the heterotrait-monotrait ratio HTMT. The next step was that after the validation of the measurement model, the structural model was tested to evaluate relationships between the constructs as hypothesized. Path coefficients, *T*-statistics, and *p*-values were calculated to probe into the significance and strength of the relationships. In parallel,

R-square values were computed to determine the explained variance in the dependent variables. Further, it examines the mediating role of teacher self-efficacy and the moderating role of pedagogical content knowledge on the relationship of digital teaching competence with faculty career well-being.

Support vector machine analysis was further used by the software JASP to strengthen the robustness and predictiveness of the results. SVM was applied to classify and predict relationships between variables, providing a nonlinear way of modeling complex interactions. The JASP has been an intuitive implementation platform with detailed performance metrics – MSE, RMSE, and R^2. Feature importance metrics were used to determine the relative contribution of key factors such as psychological influences, AI exposure, and privacy assurances to the predictive model. This integration of SVM analysis complements the PLS-SEM approach by providing a more holistic understanding of the relationships in the dataset (Anwar et al., 2025).

3 Results and discussion

3.1 Reliability and validity statistics

Tab. 2 depicts the reliability and validity statistics of the variables that were examined in depth. Tab. 2 tabulates the Cronbach's alpha value of the constructs concerned, composite reliability (CR), and average variance extracted (AVE). Regarding digital teaching competence, Cronbach's alpha is 0.889, and CR is 0.905, thus measuring exceptionally high constructs' internal consistency and reliability. The AVE of 0.661 presents a great capture of the variance in DTC by the underlying items; hence, it satisfies the threshold often accepted at 0.5 for convergent validity.

Staff career well-being also shows good reliability with Cronbach's alpha at 0.847 and composite reliability of 0.892, apart from an AVE of 0.623, proving it to be valid and implying one may possess a high level of correlation among its measurement items. Comparatively lower reliability also lies in pedagogical content knowledge with Cronbach's alpha of 0.809 and composite reliability of 0.862. Yet, the AVE of 0.513 is above the minimum threshold for convergent validity. Lastly, teacher self-efficacy (TSE) shows great reliability with Cronbach's alpha of 0.940 and composite reliability of 0.949, and an AVE of 0.591. It is great to observe such high reliability: responses to nearly all items are stable in terms of their correlation with teacher self-efficacy, which supports that the items capture the construct.

3.2 Measurement items' fitness statistics

Tab. 3 presents measurement items' fit statistics relative to the constructs relevant to each in this research. The loadings of every item indicate how much they contribute to the constructs, with the higher values indicating higher relationships. It is apparent that digital teaching competence items DTC1, DTC3, and DTC5 have high loadings of 0.818, 0.875, and 0.938, respectively, which indicate a high contribution towards this construct. Conversely, DTC2 loads low at 0.606 and thus is not thought to make a significant contribution towards overall constructs and warrants further exploration. Items measuring the career well-being of the faculty, FCW4 (0.721) and FCW8 (0.890), reflect a high representation of this construct, and other items in FCW5 (0.821) and FCW6 (0.765) also reflect high contributions. From the PCK factor, items' loading also differed from item to item with varying levels of significance strength. The strongest is PCK3 with 0.841 value, followed by PCK1

Tab. 2: Variables reliability and validity

	Cronbach's alpha	Composite reliability	Average variance extracted (AVE)
Digital teaching competence	0.889	0.905	0.661
Faculty career well-being	0.847	0.892	0.623
Pedagogical content knowledge	0.809	0.862	0.513
Teacher self-efficacy	0.940	0.949	0.591

Source: own

at 0.753. But PCK4 shows relatively weaker loading of 0.616; thus, this will weaken the overall strength of this construct. Teacher self-efficacy demonstrates a more stable performance, with loadings predominantly above 0.700, and this means that items such as TSE11 (0.882) and TSE8 (0.845) are contributing significantly. The variation in the range of loadings for the different constructs captures that most items are well performing in their categories. There are some items, such as DTC2 and PCK4, and others, such as DPC3 or TPK10, that need to be re-evaluated to enhance the overall measurement model. This analysis validates that the selected items accurately measure the constructs of interest and hence validates the study.

Tab. 3: Measurement items fitness statistics

	Digital teaching competence	*Faculty career well-being*	*Pedagogical content knowledge*	*Teacher self-efficacy*
DTC1	0.818			
DTC2	0.606			
DTC3	0.875			
DTC4	0.789			
DTC5	0.938			
FCW4		0.721		
FCW5		0.821		
FCW6		0.765		
FCW7		0.738		
FCW8		0.890		
PCK1			0.753	
PCK2			0.757	
PCK3			0.841	
PCK4			0.616	
PCK5			0.667	
PCK6			0.640	
TSE1				0.681
TSE10				0.804
TSE11				0.882
TSE12				0.829
TSE13				0.786
TSE2				0.735
TSE3				0.600
TSE4				0.712
TSE5				0.843
TSE6				0.624
TSE7				0.713
TSE8				0.845
TSE9				0.869

Note: DTC – digital teaching competence; FCW – faculty career well-being; PCK – pedagogical content knowledge; TSE – teacher self-efficacy.

Source: own

3.3 Fornell-Larcker criterion for discriminant validity

Tab. 4 shows the Fornell-Larcker criterion for determining discriminant validity amongst the study constructs. The diagonal elements are the square root of the AVE for each construct, while off-diagonal elements represent the correlation between a pair of constructs. For DTC, the square root of AVE is 0.813, which suggests that it has good discriminant validity because it is above its correlations with other constructs. The same applies to faculty career well-being, which yields an AVE value of 0.789, again above the correlations it has with other constructs, affirming the distinctiveness of the two constructs. The correlation between DTC and FCW is relatively low at 0.183, indicates that although there may be some relationship, they are distinct constructs. Pedagogical content knowledge (PCK) has a square root of AVE of 0.717. This would sustain its discriminant validity since it is higher than its correlations with other constructs, especially with FCW at 0.532 and teacher self-efficacy at 0.319. Teacher self-efficacy has an AVE of 0.768. Its correlations with the two foregoing constructs are 0.792 with FCW and 0.842 with PCK. Though TSE has a more significant correlation with these constructs, the square root of AVE of TSE remains greater than these correlation values, thereby showing that it maintains its discriminant validity. In summary, by the Fornell-Larcker criterion, all constructs depict the value of being different and apart from one another, which adds further validation to the measurement model used in this study.

3.4 Heterotrait-monotrait ratio (HTMT)

Tab. 5 reports the heterotrait-monotrait ratio (HTMT) as a supplement for the discriminant validity of the constructs. For this table, lower values signify that discriminative validity is more appropriate, and values close to or higher than 0.85 signify it could be too high. The HTMT values of digital teaching competence about other constructs are low (the maximum is 0.165 with pedagogical content knowledge and 0.138 with faculty career well-being), which also provides clear differentiation between DTC and other variables. These values suggest that DTC is distinct from other constructs in this model. For FCW, the HTMT value with TSE is 0.842 and close to a commonly referenced threshold of 0.85. This indicates a robust relationship between the two constructs. Still, this does not eliminate discriminant validity; the HTMT ratio can show distinctiveness if viewed in conjunction with other validity measures. The HTMT

Tab. 4: Fornell-Larcker criterion

Variables	DTC	FCW	PCK	TSE
1. *Digital teaching competence*	0.813			
2. *Faculty career well-being*	0.183	0.789		
3. *Pedagogical content knowledge*	0.193	0.532	0.717	
4. *Teacher self-efficacy*	0.069	0.792	0.319	0.768

Source: own

Tab. 5: HTMT

Variables	DTC	FCW	PCK	TSE
1. *Digital teaching competence*				
2. *Faculty career well-being*	0.138			
3. *Pedagogical content knowledge*	0.165	0.632		
4. *Teacher self-efficacy*	0.102	0.842	0.384	

Source: own

value between PCK and TSE is at 0.384, which further supports the conclusion that, although related, the constructs used are reasonably distinct. Overall, the HTMT results reinforce the findings from the Fornell-Larcker criterion in that the constructs used in this research are adequately differentiated and legitimate in their inclusion in the model.

3.5 *R*-square statistics

Tab. 6 summarizes the *R*-square statistics and goodness-of-fit model statistics that serve to indicate to what degree constructs in the structural model are predicted. The values of *R*-square, thus, indicate the variance explained by the independent variable as a percentage of each dependent variable in the model. *R*-square values are also given for faculty career well-being with a strong value of 0.723, which means that the model can explain approximately 72.3% of the variation of faculty career well-being and hence shows that the predictor variables have a significant impact. Conversely, the *R*-square value for teacher self-efficacy stands at 0.115, where only 11.5% of its variance is explained by the independent variables in the model. Such an unequal variance explained by the model suggests that while faculty career well-being highly determines digital teaching competence, the predictors may explain just a limited amount of variance in teacher self-efficacy. Adjusted *R*-square estimates for FCW and TSE are 0.719 and 0.112, respectively. This reflects that the model still has robust predictive capability after adjusting for the number of predictor variables included in it. As the Q^2 predict is estimated at 0.066, the model holds predictive significance with respect to FCW since a change in the variable can be adequately predicted due to a change in the predictor variables. For TSE, no Q^2 predict value is given; this could be predictive irrelevance as predictors were unable to explain a minimum variance. The *R*-square statistics tend to be high, thus indicating good explanatory ability for FCW and indicating where additional research would be needed to elucidate other factors that make up TSE.

Tab. 6: *R*-square statistics model goodness of fit statistics

Variables	*R*-square	*R*-square adjusted	Q^2 predict	RMSE	MAE
Faculty career well-being	0.723	0.719	0.642	0.066	0.072
Teacher self-efficacy	0.115	0.112			

Source: own

3.6 *F*-square statistics

Tab. 7 summarizes the use of the *F*-square statistics in reporting effect sizes, indicating the effect of independent variables on the dependent variables along the model in any analysis. Such an effect size would be small, medium, or large at more than 0.15, 0.35, and 0.50, respectively. For faculty career well-being, the *F*-square value for digital teaching competence is 0.021, indicating a small effect size. This would mean that, although some contribution to variance in FCW is due to digital teaching competence, its effect magnitude is relatively modest. By contrast, PCK had a much more significant effect on FCW with an *F*-square value of 0.286, denoting a medium effect size. This means that PCK takes a more significant role in the career well-being of faculty, further emphasizing content knowledge besides digital skills. On the other hand, TSE, one of the elements, reveals that digital teaching competence has a more pronounced effect, the *F*-square value of 0.135 when interpreted as having a medium effect size. This would mean that increases in teacher self-efficacy are bound to result from increases in digital teaching competence. The *F*-square statistics do not explicitly report on the moderating effect of pedagogical content knowledge on TSE, which calls for further exploration in this area. Besides, the *F*-square value for TSE is sufficiently large at 1.552, which suggests the effect size is large, further indicating that the variations in self-efficacy are significantly influenced by other predictors within the model.

Tab. 7: ***F*-square statistics**

Variables	*Faculty career well-being*	*Teacher self-efficacy*
Digital teaching competence	0.021	0.135
Moderating effect of pedagogical content knowledge	0.018	
Pedagogical content knowledge	0.286	
Teacher self-efficacy	1.552	

Source: own

3.7 Path analysis results

Tab. 8 and Fig. 2 present the results of the path analysis, which examines the interactions among the main variables of this research. The 0.579 path coefficient, with *T*-statistic and *p*-value 7.906 and 0.000, respectively, indicates that digital teaching competence has a significant influence on faculty career well-being. Hence, this robust positive association means that the greater the development of digital teaching capabilities of a faculty member, the more his/her career overall well-being will be greatly influenced too. Additionally, it was also noted that teacher self-efficacy is heavily influenced by digital teaching competence; the path coefficient was calculated to be 0.369 and, the *T*-statistic was calculated as 4.855 and *p*-value = 0.000. This means that academic staff with digital teaching competencies have greater faith in their teaching.

In addition, the outcome also signifies an indirect positive relationship between digital teaching competencies and faculty career well-being with the mediator of teachers' self-efficacy, such that path coefficient = 0.448 and *T*-statistic = 6.816, *p*-value = 0.000. This has emphasized the role of teacher self-efficacy as a vital mediator, through which higher self-efficacy enhances the positive effect of digital competence on faculty well-being.

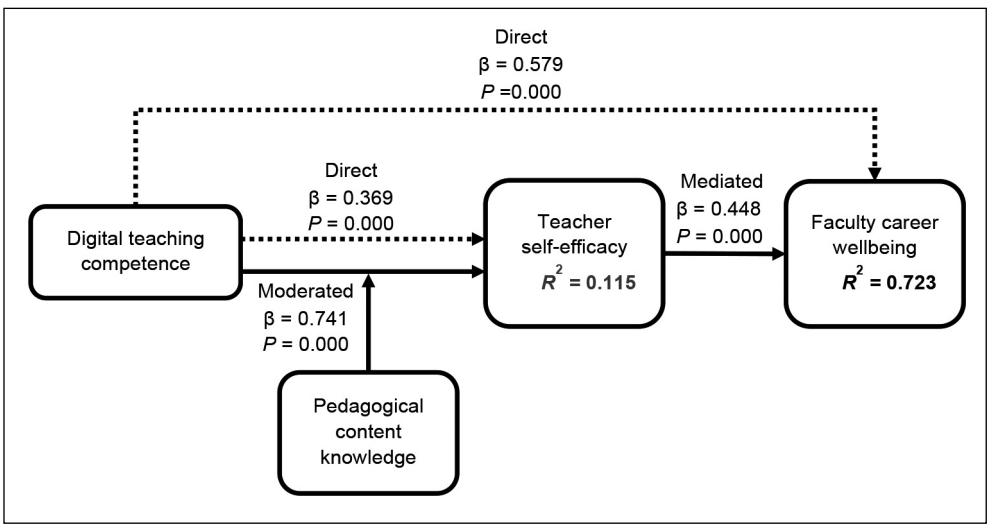


Fig. 2: **Structural model for path analysis**

Source: own

Tab. 8: Path analysis

Path analyses	Original sample	Sample mean	Standard deviation	*T*-statistics	*p*-values
Digital teaching competence significantly influences the faculty's career well-being.	0.579	0.687	0.087	7.906	0.000
Digital teaching competence significantly influences teacher self-efficacy.	0.369	0.475	0.080	4.855	0.000
Teacher self-efficacy significantly mediates the relationship between digital teaching competence and the faculty's career well-being.	0.448	0.454	0.059	6.816	0.000
Pedagogical content knowledge significantly moderates the relationship between digital teaching competence and teacher self-efficacy.	0.741	0.746	0.040	2.042	0.000

Source: own

Finally, the connection between digital teaching proficiency and teacher self-efficacy is greatly moderated by pedagogical content knowledge, as evidenced by the path coefficient of 0.741 and the *T*-statistic of 2.042 with *p*-value = 0.000. This would suggest that good pedagogical content knowledge reinforces the positive effects of digital teaching proficiency on faculty well-being thus it is a critical moderating variable. These results emphasize the interaction among digital teaching competence, self-efficacy, and pedagogical knowledge for promoting faculty well-being.

3.8 Performance metrics of the model

Tab. 9 gives the performance metrics of the model, which show the MSE (mean squared error) = 0.006. This means the average squared error between predicted values and actual values is 0.006. MSE scaled version of (MSE scaled) is 0.244. Therefore, this means that the value of the MSE is scaled to a range of 0.244. The RMSE (root mean squared error) is equal to 0.077. The square root of MSE measures the average difference between the predicted and actual values. MAE/MAD (mean absolute error/mean absolute deviation) is 0.064. This is the average absolute difference between the predicted and actual values. MAPE (mean absolute percentage error) is 13.72%. This is the average absolute percentage error between the predicted and actual values. *R*-squared (R^2) is 0.728, this implies that the model explained 72.8% of the change in the dependent variable.

3.9 Predictive power or performance plot

Fig. 3 displays the predictive power or performance plot of the model. The predicted values

Tab. 9: Model performance metrics

Performance metrics	Value
MSE	0.006
MSE (scaled)	0.244
RMSE	0.077
MAE/MAD	0.064
MAPE (%)	13.720
R^2	0.728

Source: own

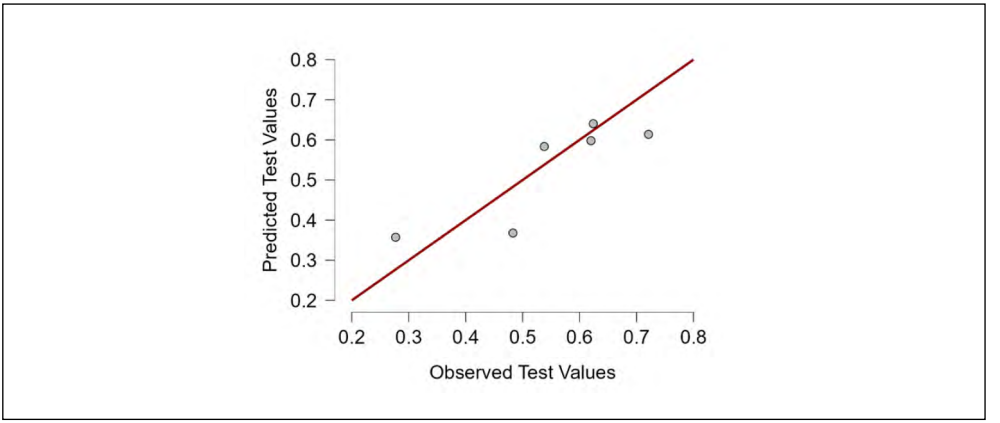


Fig. 3: **Predictive performance plot**

Source: own

are on the y-axis, while the observed values are plotted on the x-axis. The diagonal represents a perfect prediction. The points in the plot display the actual predictions made by the model. The distance of these points from the diagonal will determine the model's accuracy. The nearer the points are to the diagonal line, the more accurate the model is to be considered.

3.10 Additive explanations for predictions

Tab. 10 demonstrates additive explanations for the predictions of test set cases and how different features influence the collective outcomes of an overall prediction (Tab. 10). The different test cases are identified in the "Case" column, while in the "Predicted" column, the predicted values accompanying each case are revealed. The "Base" column indicates the baseline prediction for each case, showing what would happen if all other features were set to zero. Tab. 10 also provides the contribution of specific factors to the predictions, with the "Digital teaching competence" column reflecting how this particular feature impacts the overall prediction for each case. Then, a column will represent the moderating effect for prediction as part of "Moderating effect 1." Also, it will show the contribution that goes to teacher self-efficacy for the overall prediction of any given case. Thus, these columns offer a comprehensive view of all kinds of interacting features and what is coming about in terms of their respective contribution to the predictive model results.

Tab. 10: **Additive explanations for predictions of test set cases**

Case	Predicted	Base	Digital teaching competence	Moderating effect 1	Teacher self-efficacy
1	0.614	0.452	0.044	0.021	0.098
2	0.368	0.452	5.755×10^{-4}	0.033	−0.117
3	0.357	0.452	0.019	0.022	−0.135
4	0.64	0.452	0.048	0.023	0.118
5	0.583	0.452	0.024	0.064	0.044

Source: own

3.11 Mean squared error plot

Fig. 4 shows the mean squared error (MSE) plot, this plot visualizes the model's performance vs. "cost of violation of constraints." X-axis: cost of violation of constraints. It describes how the model can violate constraints imposed during the training process. The higher cost means there is more freedom for deviation. Y-axis: mean square error. It measures how much the difference between what the model predicted and the value is on average. The better the performance, the lower the mean squared error. Dashed line: training set mean squared error. This measures how well a model does on data that it already saw in training. Solid line: validation set mean squared error. This measures how well the model performs on entirely new, previously unseen data.

Performance trend: as the cost of violation of constraints increases (moving right along the X-axis), the mean squared error for both training and validation sets generally decreases. Allowing the model more freedom to violate constraints leads to better performance. Overfitting: the dashed line (training set) is always lower than the solid line (validation set), which means the model fits the training data too closely. This means the model has overlearned training data patterns, generalizing less accurately to unseen data. The plot describes a trade-off between the constraint violation and the model's performance. Higher flexibility could improve overall performance, but there is always the possibility of overfitting that reduces a model's ability to generalize well to new data. Balancing the violation of the constraint

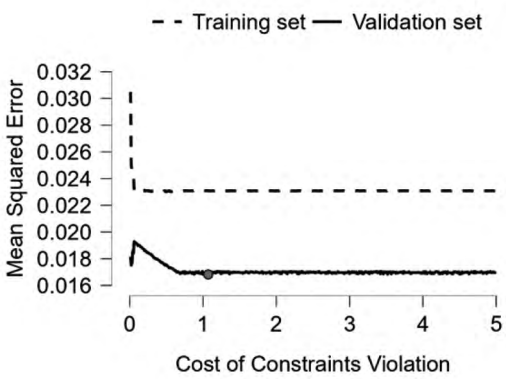


Fig. 4: Mean squared error plot

Source: own

with model complexity is an important aspect of building robust and reliable models.

4 Discussion

The findings of this research are highly insightful in grasping the intricate interplays among digital teaching ability, teacher self-efficacy, and staff career well-being. Since education is now on its way to going digital, it becomes of immense importance to know how these constructs interact and dance around one another in order to improve effective pedagogical techniques further and sustain academic settings. This discussion chapter outlines the implications of the accepted hypotheses, such as empirical results, to be compared with current literature and the theoretical frameworks. Through an analysis of the accepted hypotheses, we can further describe why enhancing the faculty member's level of digital teaching competencies affects self-efficacy and overall career happiness and wellness.

Confirmation of the first hypothesis – that digital competence has a strong impact on faculty career well-being should reinforce increasing consensus in education research that technology is key to securing positive faculty experiences. Findings indicate that teachers with high levels of digital competency are best placed to deal with challenges from new education practices, giving rise to even greater job satisfaction and higher work engagement. This is consistent with earlier research that has shown that it is through the proper incorporation of technology in instruction that teachers can have more sense of involvement and satisfaction (Kholifah et al., 2023; Ninković et al., 2024). Digital competencies promote the abilities of instructors towards more active learning, professional competence, and satisfaction. The findings highlight that effective support for institutional and professional development on digital matters can only be initiated if the need for training and resources in this respect is ongoing and related to the increasing digitization of the environment.

The second supported hypothesis also posits that digital instructional ability affects teacher self-efficacy significantly; therefore, with appropriate technology, teachers' self-confidence in teaching will increase. It shows that the teachers who think that they are proficient and capable of digital teaching are likely to think that they are excellent teachers; these two also create a positive correlation with general self-efficacy (Wong et al., 2023). Since this is a significant relationship, greater self-efficacy has been linked to enhanced motivation, more impressive persistence under adversity, and more effective instructional practices. When teachers have confidence in their abilities, they will attempt new, more effective instructional practices and will engage meaningfully with students in class, leading to an active learning environment for everyone. This study emphasizes the need for higher education institutions to incorporate digital competencies into the pedagogy design process of the faculty of a strategic academic approach. This implies that self-efficacy and job satisfaction in general become essential to individual teachers and the educational community.

The reception of hypothesis three shows that teacher self-efficacy deeply mediates the relationship between digital teaching competence and faculty career well-being. This finding emphasizes the role of self-efficacy as a significant psychological mechanism that makes digital teaching competencies consequential for well-being. The results yield the role of self-efficacy as a mediator because it promotes enhancing digital skills and beliefs in one's ability to use digital skills effectively (Uçar & Zarfsaz, 2023). Educators with a higher self-efficacy tend to approach their teaching roles with optimism, resilience, and a better end towards job satisfaction and more commitment to the profession. Therefore, this suggests a need for educational institutions to develop training that develops self-efficacy through mentoring, collaborative skills, and positive feedback, creating a supportive context for developing skills and psychological empowerment.

The fourth hypothesis acceptance ensures that pedagogical content knowledge strongly moderates digital teaching competence and teacher self-efficacy. This identifies the sophistication of knowledge fields, in which digital competence is more effective with the presence of robust pedagogical content knowledge. Teachers with extensive subject knowledge and sound teaching practices are capable of effectively incorporating digital technologies for improving teaching and well-being (Göbel et al., 2023). This highlights the need for professional development that involves both pedagogical and digital skills, facilitating more effective teaching by faculty. Institutions need to create collaborative learning environments where the faculty can exchange best practices to improve job satisfaction and professional development (Khan et al., 2022).

In all, the findings illustrate the interconnectedness of teacher self-efficacy, digital teaching competence, and faculty career well-being. The endorsement of all four hypotheses underscores the importance of enhancing digital competence and self-efficacy among teachers since these directly impact their job satisfaction and teaching experience. Identifying pedagogical content knowledge as a significant moderator, schools need to prepare faculty to deal with contemporary teaching issues. Future initiatives should aim to develop digital competence, self-efficacy, and pedagogical knowledge to promote teacher well-being and enhance student learning.

4.1 Implications of the research

Theoretically, this research makes a contribution to the literature by bridging Bandura's social cognitive theory and the technology acceptance

model to account for the interplay between digital teaching capability, self-efficacy, and the career well-being of faculty members. The results affirm the centrality of digital competency in influencing faculty lives and point to the moderating effect of pedagogical content knowledge. By showing these high-order interplays, this research builds on current frameworks and offers groundwork for future inquiries into faculty development within digital learning.

Practically speaking, the research highlights the imperative for institutions to advance faculty digital capabilities through selective professional development opportunities. Professional development efforts should not only address digital capabilities but also prioritize pedagogical approaches to maximizing technology in teaching. Additionally, creating an environment of collaborative learning where best practices are exchanged among faculty members can improve job satisfaction and instruction. Institutional policies must focus on ongoing support for faculty in the integration of digital tools to realize long-term gains for both students and educators.

4.2 Limitations and future research directions

This research has some limitations that must be noted. The study was carried out in Saudi Arabian colleges, which can restrict the external validity of the findings to other education systems with varying technological support and pedagogical approaches. The study was also cross-sectional in nature, which fails to reflect changes in faculty digital competency and well-being over time. Longitudinal studies might yield more insights into the changing effect of digital teaching competence on the career well-being of faculty members.

Future studies ought to investigate qualitative approaches to develop a better insight into faculty use of digital tools and their effect on teaching methodologies. Broadening the study across educational levels and disciplines would better establish the validity of the findings and increase their generalizability. Further, exploring other prospective moderators, like institutional support and technology accessibility, may offer a more detailed view of the drivers of faculty career well-being in digital education.

Conclusions

Finally, this study inquiry is compelling for interactive interactions among digital instructional ability, teaching staff career health, instructor self-confidence, and instructional content information. The inference of this research sustains the premises that augmenting teachers' teaching technology/digital abilities may powerfully help them feel effective and satisfied at work overall, indicating the integral part professional training may have to contribute in today's educational contexts. To wrap it up, after acknowledging the moderating and mediating effects of pedagogical content knowledge and self-efficacy, the study opines that technological and pedagogical expertise needs to be blended into faculty training. With the dynamic digital ecosystem changing every minute, responding to schools' changing needs is the key to which this research lays on its foundation for informing decision and policy implementation regarding an efficient strategy for teaching well-being and thus having implications for student outcomes.

This research adds to the literature on digital teaching competence, career well-being of faculty, and teacher self-efficacy as developed in the knowledge base that guides educators by shedding light on inter-relationships between these constructs. Since all the hypotheses were accepted, it emphasizes the integration of digital competencies into educational psychology and professional development framework. Findings indicate that digital teaching capability directly influences career well-being among faculty while it influences positively teachers' self-efficacy as a mediator to this connection. The findings, therefore, support the self-efficacy theory that might advocate for enhancing emotions of competence and satisfaction regarding jobs through improving digital capacity among academics. Notably, the pedagogical content knowledge moderating effect directs to the mastery by teachers of both content and technology in creating maximal professional results. This way, such concepts enhance a collection of theoretical models in educational research, particularly when discussing teacher development; this would encourage researchers to investigate further relationships between technology, pedagogy, and teachers' well-being.

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